LECTURE NOTES

Lecture 5:

AI504 - Knowledge Represntation

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Contents

Recall

Recall the logic of "all p are q"

Now fix an underlying set P

- a P-sentence is just a phrase "all $\underbrace{\text{all } p \text{ are } q}_{(p,q)\text{-mathematically}}$, where $p, q \in P$"
- a P-model is a set M and a function $P \to \mathcal{P}(M)$, if $p \in P$, them $[\![p]\!] \subset P$

Terminology

P is the <u>signature</u> of a P-model or P-sentence

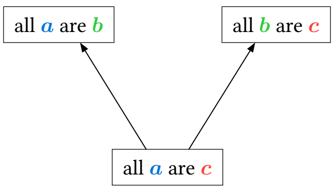
Suppose $\mathcal{M}$ is a P-model and "all p are q" is P-sentense

Then $\mathcal{M} \vDash \text{all } p \text{ are } q$ means $[\![p]\!] \subset [\![q]\!]$

Suppose Γ is a subset of senteces and ϕ is a single sentences, them $\Gamma \vDash \phi$ mean that for all models $\mathcal{M}$, if $\mathcal{M} \vDash \Gamma$, then $M \vDash \phi$

Prooftrees

A <u>proof tree</u> is a binary tree labeled by sentences such that at each non-leaf node



Suppose that T is a proof tree. Then T is a **proof** of ϕ from Γ if:

1. ϕ labels the root of T
2. every leaf of T either is in Γ or has the form $\underbrace{\text{all \_ are \_}}_{\text{blanks are equal}}$

Definition: $\Gamma \vdash \phi$

$\Gamma \vdash \phi$ if there is a proof tree of ϕ from Γ

During this and the next lecture we want to prove

$$\Gamma \vDash \phi \iff \Gamma \vdash \phi$$

where

$$\underbrace{\Gamma \vDash \phi \implies \Gamma \vdash \phi}_{\text{Soundness } (\vDash \text{ implies } \vdash)}$$

and

$$\underbrace{\Gamma \vDash \phi \impliedby \Gamma \vdash \phi}_{\text{Completeness } (\vdash \text{ implies } \vDash)}$$

Soundness

Definition: Soundness

For any set of sentences Γ and sentence ϕ, if $\Gamma \vdash \phi$ then $\Gamma \vDash \phi$

For all Γ, ϕ: if there exists a proof of ϕ from Γ, then for every model $\mathcal{M}$, if $\mathcal{M} \vDash \Gamma$ then $\mathcal{M} \vDash \phi$.

Message

for all proofs T, Γ and ϕ.

If T is a proof of ϕ from Γ then for every model $\mathcal{M}$,

if $M \vDash \Gamma$ then $\Gamma \vDash \phi$

Now we can do induction to prove this, where goal

if T is a proof of ϕ from Γ and $\mathcal{M}$ is any model satisfying Γ them $\mathcal{M} \vDash \phi$ too

Proof by induction

Base Case) (Assume proof T is a single leaf)

Assume proof T is a single leaf

Then ϕ is the only thing in T, since T is a proof from ϕ.

Notice that all the leaves from T fomrs from Γ

Let $\mathcal{M}$ be any model that satisfies (all sentences in) Γ

since $\phi \in \Gamma$, $\mathcal{M} \vDash \phi$

Inductive step

Suppose that T is not a leaf

Decompose T as follows. Say that ϕ is "all p are q"

Then there exists an r s.t T looks like

$$\begin{tikzcd}
\boxed{\text{all } p \text{ are } r} & & \boxed{\text{all } r \text{ are } q} \\
& \boxed{\text{all } p \text{ are } q} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

then let T_0, T_1 be the left & right havles of T

T_0 is a proof of "all p are r" from Γ (**IH** but explain better)

T_1 is a proof of "all r are q" from Γ (**IH** but explain better)

Suppose that $\mathcal{M} \vDash \Gamma$.

By **IH** applied to T_0, $\mathcal{M} \vDash$ all p are r

By **IH** applied to T_0, $\mathcal{M} \vDash$ all p are r

Since $\mathcal{M} \vDash$ all p are r

$[\![p]\!] \subset [\![r]\!]$

Since $\mathcal{M} \vDash$ all r are q

$[\![r]\!] \subset [\![q]\!]$

Therefore,

$$[\![p]\!] \subset [\![q]\!], \text{so } \mathcal{M} \vDash \text{all } p \text{ are } q$$